

6 (a) Table 6.1 compares some of the features of DNA with the protein collagen.

Complete Table 6.1.

Table 6.1

feature	DNA	collagen
elements		C H O N
monomers		
bond between monomers		
site of production in eukaryotic cells	nucleus	

[6]

(b) Most of the DNA in a eukaryotic cell is located in the nucleus.

State how the structure of a nucleus is suited to its function of containing DNA.

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..... [2]

(c) Fibroblast cells produce elastic fibres and collagen fibres in lung tissue.

Describe the function of elastic fibres in lung tissue.

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..... [2]

(d) Name the stage of the cell cycle during which DNA **and** proteins are synthesised.

..... [1]

[Total: 11]